

Lab 6: Update Sequence Number (USN) Journaling

What You Need for this lab

- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>
 - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- https://www.dropbox.com/sh/3lyzpi7vyod0pfs/AADh2b3TnmMqWZHB1Uzv9_ma?dl=0
- https://ubaltmy.sharepoint.com/:f/g/personal/id31ga53_ubalt_edu/EoJThlUPE85Hm7592HB5sjlBGo4Ht9drLxt9XnwDdnf3PA?e=4Okj4m
- <https://github.com/jschicht/ExtractUsnJrnl>
- Image .dd from Lab 5

Scenario 1

- Install *UsnJrnl2Csv* tool

```
hainguyen@hainguyen: ~/Lab
$ ls -l
total 26272196
-rw-rw-r-- 1 hainguyen hainguyen 74242 Jun 15 04:40 'AADh2b3TnmMqWZHB1Uzv9_ma?dl=0'
-rw-rw-r-- 1 hainguyen hainguyen 2147483648 Jun 15 04:19 cfreds_2015_data_leakage_pc.7z.001
-rw-rw-r-- 1 hainguyen hainguyen 2147483648 Jun 15 04:26 cfreds_2015_data_leakage_pc.7z.002
-rw-rw-r-- 1 hainguyen hainguyen 1132827932 Jun 15 04:32 cfreds_2015_data_leakage_pc.7z.003
-rw-rw-r-- 1 hainguyen hainguyen 21474836480 Apr 21 2015 cfreds_2015_data_leakage_pc.dd

(hainguyen@hainguyen)~/Lab
$ git clone https://github.com/jschicht/UsnJrnl2Csv.git
Cloning into 'UsnJrnl2Csv'...
remote: Enumerating objects: 182, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 182 (delta 0), reused 2 (delta 0), pack-reused 176 (from 1)
Receiving objects: 100% (182/182), 12.96 MiB | 7.52 MiB/s, done.
Resolving deltas: 100% (111/111), done.

(hainguyen@hainguyen)~/Lab
$ icat -i raw -o 206848 cfreds_2015_data_leakage_pc.dd 59016-128-3 > UsnJrnl2Csv/usnJrnl.bin

(hainguyen@hainguyen)~/Lab
$ ls UsnJrnl2Csv/usnJrnl.bin -l
-rw-rw-r-- 1 hainguyen hainguyen 69767168 Jun 15 04:44 UsnJrnl2Csv/usnJrnl.bin

(hainguyen@hainguyen)~/Lab
$
```

- Run *UsnJrnl2.exe* to covert *UsnJrnl.bin* to .csv

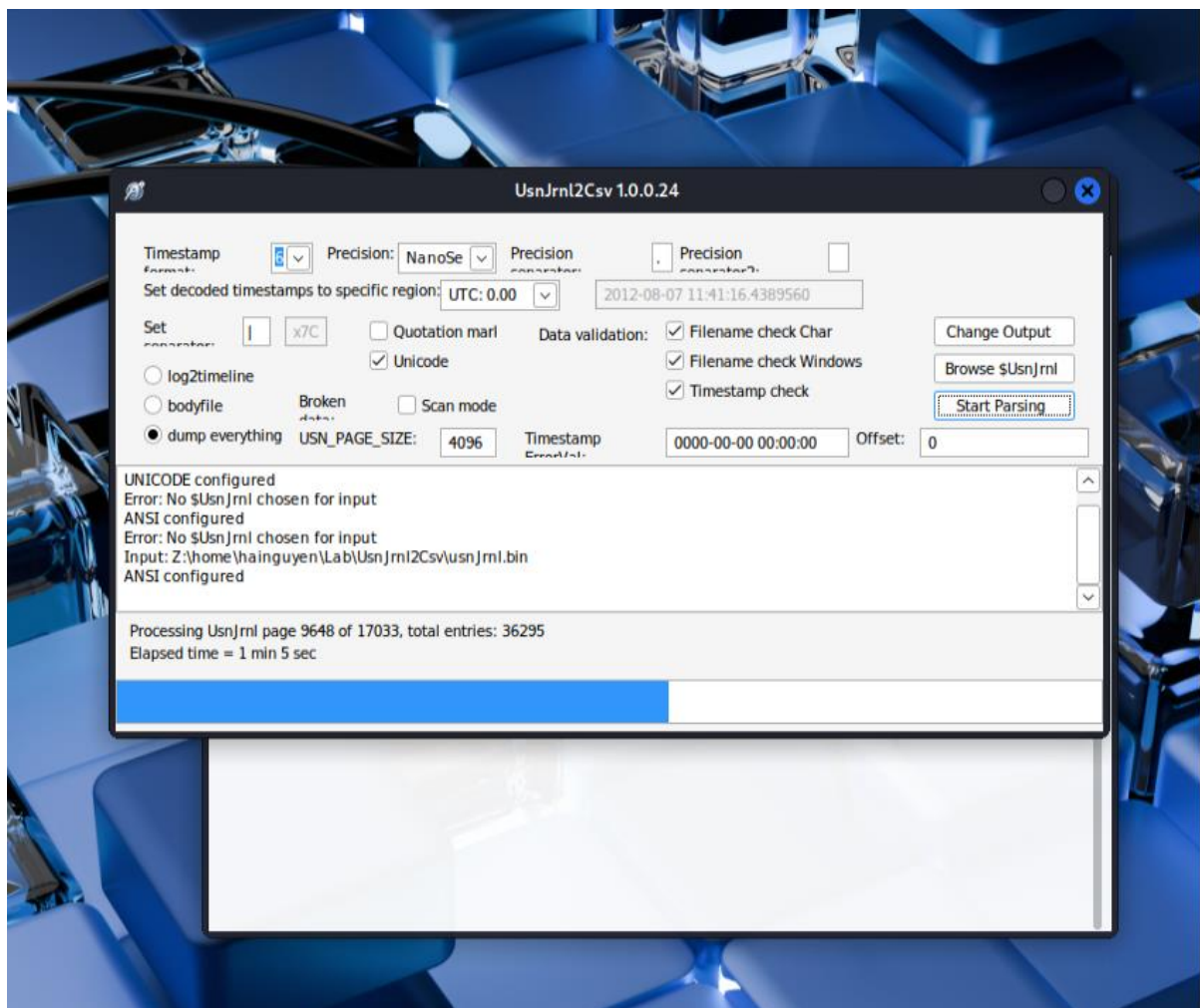
```
hainguyen@hainguyen: ~/Lab/UsnJrnl2Csv
File Actions Edit View Help

(hainguyen@hainguyen)~]
$ cd Lab

(hainguyen@hainguyen)~/Lab]
$ cd UsnJrnl2Csv

(hainguyen@hainguyen)~/Lab/UsnJrnl2Csv]
$ wine UsnJrnl2Csv64.exe
```

- Set configuration: Browser to *usnJrnl.bin*



- Show the extracted USN journal file

```

Kali- Linux (Basic) [Running] - Oracle VM VirtualBox
File Actions Edit View Help

(hainguyen@hainguyen)-[~]
$ cd Lab

(hainguyen@hainguyen)-[~/Lab]
$ cd UsnJrnl2Csv

(hainguyen@hainguyen)-[~/Lab/UsnJrnl2Csv]
$ ls Usn* -l
-rw-rw-r-- 1 hainguyen hainguyen      3 Jun 15 05:15 UsnJrnl_2025-06-15_05-15-37.csv
-rw-rw-r-- 1 hainguyen hainguyen     40 Jun 15 05:15 UsnJrnl_2025-06-15_05-15-37.log
-rw-rw-r-- 1 hainguyen hainguyen    939 Jun 15 05:15 UsnJrnl_2025-06-15_05-15-37.sql
-rw-rw-r-- 1 hainguyen hainguyen      0 Jun 15 05:16 UsnJrnl_2025-06-15_05-16-17.csv
-rw-rw-r-- 1 hainguyen hainguyen     37 Jun 15 05:16 UsnJrnl_2025-06-15_05-16-17.log
-rw-rw-r-- 1 hainguyen hainguyen    941 Jun 15 05:16 UsnJrnl_2025-06-15_05-16-17.sql
-rw-rw-r-- 1 hainguyen hainguyen 46649991 Jun 15 05:28 UsnJrnl_2025-06-15_05-19-06.csv
-rw-rw-r-- 1 hainguyen hainguyen    490 Jun 15 05:28 UsnJrnl_2025-06-15_05-19-06.log
-rw-rw-r-- 1 hainguyen hainguyen    928 Jun 15 05:19 UsnJrnl_2025-06-15_05-19-06.sql
-rw-rw-r-- 1 hainguyen hainguyen 1067008 Jun 15 05:08 UsnJrnl2Csv64.exe
-rw-rw-r-- 1 hainguyen hainguyen   64380 Jun 15 04:43 UsnJrnl2Csv.au3

(hainguyen@hainguyen)-[~/Lab/UsnJrnl2Csv]
$ cat UsnJrnl_2025-06-15_05-36-53.csv | head -n 2
Offset|FileName|USN|Timestamp|Reason|MFTReference|MFTReferenceSeqNo|MFTParentReference|MFTParentReferenceSeqNo|FileAttributes|MajorVersion|MinorVersion|SourceInfo|SecurityId
0x01F80000|api-ms-win-core-fibers-l1-1-0.dll|33030144|2015-03-22 15:15:08.2192456|BASIC_INFO_CHANGE+CLOSE+INDEXABLE_CHANGE|63536|4|63453|4|archive|2|0|0x00000000|0
00000|0

(hainguyen@hainguyen)-[~/Lab/UsnJrnl2Csv]
$

```

- Only show “RENAME_OLD” information of any .jpg files

```

(hainguyen@hainguyen)-[~/Lab/UsnJrnl2Csv]
$ grep -P "\.jpg.*RENAME_OLD" UsnJrnl_2025-06-15_05-36-53.csv --color
0x03D6BCA0|Chrysanthemum.jpg|64404640|2015-03-24 20:11:42.8361742|RENAME_OLD_NAME|74620|285|21921|3|archive|2|0|0x00000000|0
0x03D6BF68|Desert.jpg|64405352|2015-03-24 20:11:42.8517742|RENAME_OLD_NAME|74625|18|21921|3|archive|2|0|0x00000000|0
0x03D6C528|Hydrangeas.jpg|64406324|2015-03-24 20:11:42.8517742|RENAME_OLD_NAME|74641|18|21921|3|archive|2|0|0x00000000|0
0x03D6C6C0|Jellyfish.jpg|64408264|2015-03-24 20:11:42.8517742|RENAME_OLD_NAME|74656|8|21921|3|archive|2|0|0x00000000|0
0x03D6C088|Koala.jpg|64408968|2015-03-24 20:11:42.8517742|RENAME_OLD_NAME|74666|10|21921|3|archive|2|0|0x00000000|0
0x03D6D058|Lighthouse.jpg|64409688|2015-03-24 20:11:42.8517742|RENAME_OLD_NAME|74667|16|21921|3|archive|2|0|0x00000000|0
0x03D6D318|Penguins.jpg|64410392|2015-03-24 20:11:42.8517742|RENAME_OLD_NAME|74670|12|21921|3|archive|2|0|0x00000000|0
0x03D6D5D8|Tulips.jpg|64411096|2015-03-24 20:11:42.8517742|RENAME_OLD_NAME|74676|10|21921|3|archive|2|0|0x00000000|0
0x0411B558|Chrysanthemum.jpg|68269408|2015-03-25 15:13:39.1698055|RENAME_OLD_NAME|74620|285|74402|9|not_indexed|2|0|0x00000000|0
0x0411C4F8|Desert.jpg|68273408|2015-03-25 15:13:39.4350855|RENAME_OLD_NAME|74625|10|74402|9|not_indexed|2|0|0x00000000|0
0x0411D268|Hydrangeas.jpg|68276848|2015-03-25 15:13:39.5442055|RENAME_OLD_NAME|74641|18|74402|9|not_indexed|2|0|0x00000000|0
0x0411F378|Jellyfish.jpg|68285808|2015-03-25 15:13:48.8730055|RENAME_OLD_NAME|74656|8|74402|9|not_indexed|2|0|0x00000000|0
0x04120428|Koala.jpg|68293668|2015-03-25 15:13:49.0602055|RENAME_OLD_NAME|74666|10|74402|9|not_indexed|2|0|0x00000000|0
0x04121188|Lighthouse.jpg|68293088|2015-03-25 15:13:49.2162855|RENAME_OLD_NAME|74667|16|74402|9|not_indexed|2|0|0x00000000|0
0x04122058|Penguins.jpg|68296792|2015-03-25 15:13:49.3566055|RENAME_OLD_NAME|74670|12|74402|9|not_indexed|2|0|0x00000000|0
0x04122EC8|Tulips.jpg|68300488|2015-03-25 15:13:49.4658055|RENAME_OLD_NAME|74676|10|74402|9|not_indexed|2|0|0x00000000|0

(hainguyen@hainguyen)-[~/Lab/UsnJrnl2Csv]
$

```

- Only show “RENAME_OLD” information of any .jpg files. Only show three columns, FileName, Timestamp, Reason

```
(hainguyen@hainguyen)~/Lab/UsnJrnl2Csv
$ awk 'BEGIN { FS="|" } { print NR, $2, $4, $5 }' UsnJrnl_2025-06-15_05-36-53.csv | grep -E "\.jpg.*RENAME_OLD" --color
261131 Chrysanthemum.jpg 2015-03-24 20:11:42.8361742 RENAME_OLD_NAME
261139 Desert.jpg 2015-03-24 20:11:42.8517742 RENAME_OLD_NAME
261155 Hydrangeas.jpg 2015-03-24 20:11:42.8517742 RENAME_OLD_NAME
261171 Jellyfish.jpg 2015-03-24 20:11:42.8517742 RENAME_OLD_NAME
261179 Koala.jpg 2015-03-24 20:11:42.8517742 RENAME_OLD_NAME
261187 Lighthouse.jpg 2015-03-24 20:11:42.8517742 RENAME_OLD_NAME
261195 Penguins.jpg 2015-03-24 20:11:42.8517742 RENAME_OLD_NAME
261203 Tulips.jpg 2015-03-24 20:11:42.8517742 RENAME_OLD_NAME
299494 Chrysanthemum.jpg 2015-03-25 15:13:39.1698055 RENAME_OLD_NAME
299536 Desert.jpg 2015-03-25 15:13:39.4350055 RENAME_OLD_NAME
299578 Hydrangeas.jpg 2015-03-25 15:13:39.5442055 RENAME_OLD_NAME
299665 Jellyfish.jpg 2015-03-25 15:13:48.8730055 RENAME_OLD_NAME
299707 Koala.jpg 2015-03-25 15:13:49.0602055 RENAME_OLD_NAME
299740 Lighthouse.jpg 2015-03-25 15:13:49.2162055 RENAME_OLD_NAME
299791 Penguins.jpg 2015-03-25 15:13:49.3566055 RENAME_OLD_NAME
299833 Tulips.jpg 2015-03-25 15:13:49.4658055 RENAME_OLD_NAME
(hainguyen@hainguyen)~/Lab/UsnJrnl2Csv
$
```

- Show what has happened around a specific time with the same MFT reference #

```
(hainguyen@hainguyen)~/Lab/UsnJrnl2Csv
$ grep "2015-03-24 20:11:42.8361742" UsnJrnl_2025-06-15_05-36-53.csv
0x03D6BCA0|Chrysanthemum.jpg|64404640|2015-03-24 20:11:42.8361742|RENAME_OLD_NAME|74620|285|21921|3|archive|2|0|0x00000000|0
0x03D6BD00|$RKXD1U3.jpg|64404736|2015-03-24 20:11:42.8361742|RENAME_NEW_NAME|74620|285|15721|2|archive|2|0|0x00000000|0
0x03D6BD58|$RKXD1U3.jpg|64404824|2015-03-24 20:11:42.8361742|CLOSE+RENAME_NEW_NAME|74620|285|15721|2|archive|2|0|0x00000000|0
0x03D6BD80|$RKXD1U3.jpg|64404912|2015-03-24 20:11:42.8361742|SECURITY_CHANGE|74620|285|15721|2|archive|2|0|0x00000000|0
0x03D6BE00|$RKXD1U3.jpg|64405000|2015-03-24 20:11:42.8361742|CLOSE+SECURITY_CHANGE|74620|285|15721|2|archive|2|0|0x00000000|0
(hainguyen@hainguyen)~/Lab/UsnJrnl2Csv
$
```

Scenario 2

- Install *usncarve* tool (note we have changed the path)

```
(venv)~(hainguyen@hainguyen)~/Lab
$ pip install usncarve
Collecting usncarve
  Downloading usncarve-1.2.2.tar.gz (2.8 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: usncarve
  Building wheel for usncarve (pyproject.toml) ... done
  Created wheel for usncarve: filename=usncarve-1.2.2-py3-none-any.whl size=2953 sha256=6c4cd3208fea973c9da0f174b699e09f8d374abb552239b1ec514336c04cbc87
  Stored in directory: /home/hainguyen/.cache/pip/wheels/e8/18/e8/5d4c1044c25c444be62657edcd09f368ac4519e61a043c74c6
Successfully built usncarve
Installing collected packages: usncarve
Successfully installed usncarve-1.2.2
(hainguyen@hainguyen)~/Lab
$
```

- Verify installation

```
(venv)~(hainguyen@hainguyen)~/Lab
$ usncarve.py -h
usage: usncarve.py [-h] -f FILE -o OUTFILE

options:
  -h, --help            show this help message and exit
  -f, --file FILE       Carve USN records from the given file
  -o, --outfile OUTFILE Output to the given file

(hainguyen@hainguyen)~/Lab
$
```

- Install usnparser tool

```
(venv)~(hainguyen@hainguyen)~[~/Lab]
$ pip install usnparser
Collecting usnparser
  Downloading usnparser-4.1.5.tar.gz (6.1 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: usnparser
  Building wheel for usnparser (pyproject.toml) ... done
  Created wheel for usnparser: filename=usnparser-4.1.5-py3-none-any.whl size=5617 sha256=ea5b119c8625c106707cb28fd8b78e4ee0b1fc4f14ab824838b908b73c435e8d
  Stored in directory: /home/hainguyen/.cache/pip/wheels/c9/8d/4d/3384811dedf2dd3fff3e61e45f5428ff8d83e6543916e5af36
Successfully built usnparser
Installing collected packages: usnparser
Successfully installed usnparser-4.1.5

(venv)~(hainguyen@hainguyen)~[~/Lab]
$
```

- Verify installation

```
(venv)~(hainguyen@hainguyen)~[~/Lab]
$ usn.py -h
usage: usn.py [-h] [-b] [-c] [-f FILE -o OUTFILE [-s SYSTEM] [-t] [-v]

options:
  -h, --help            show this help message and exit
  -b, --body            Return USN records in comma-separated format
  -c, --csv            Return USN records in comma-separated format
  -f, --file FILE      Parse the given USN journal file
  -o, --outfile OUTFILE
                        Parse the given USN journal file
  -s, --system SYSTEM  System name (use with -t)
  -t, --tln            TLN ou2tput (use with -s)
  -v, --verbose        Return all USN properties for each record (JSON)

(venv)~(hainguyen@hainguyen)~[~/Lab]
$
```

- *usncarve.py* -f input file -o output file

```
(venv)~(hainguyen@hainguyen)~[~/Lab]
$ usncarve.py -f cfreds_2015_data_leakage_pc.dd -o usn.raw

(venv)~(hainguyen@hainguyen)~[~/Lab]
$ ls -l usn.raw
-rw-rw-r-- 1 hainguyen hainguyen 46465397 Jun 15 06:08 usn.raw

(venv)~(hainguyen@hainguyen)~[~/Lab]
$
```

- `usn.py -f input file -o output file`

```
(venv)-(hainguyen@hainguyen)-[~/Lab]
$ usn.py -f usn.raw -o usn.csv
/home/hainguyen/venv/bin/usn.py:139: DeprecationWarning: datetime.datetime.
utcfromtimestamp() is deprecated and scheduled for removal in a future versio
n. Use timezone-aware objects to represent datetimes in UTC: datetime.dateti
me.fromtimestamp(timestamp, datetime.UTC).
    return str(datetime.datetime.utcnow().timestamp() * 1e-7 - 11644473600)
)

(venv)-(hainguyen@hainguyen)-[~/Lab]
$ ls -l usn.csv
-rw-rw-r-- 1 hainguyen hainguyen 40900041 Jun 15 06:09 usn.csv

(venv)-(hainguyen@hainguyen)-[~/Lab]
$
```

- View renamed files

```
(venv)-(hainguyen@hainguyen)-[~/Lab]
$ cat usn.csv | grep -i rename | head
grep: (standard input): binary file matches
2015-03-22 15:14:17.097954 | edb0043D.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-22 15:14:17.097954 | edbtmp.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_NEW_NAME
2015-03-22 15:14:17.097954 | edbtmp.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_NEW_NAME CLOSE
2015-03-22 15:14:17.097954 | edb.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-22 15:14:17.097954 | edb0044E.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_NEW_NAME
2015-03-22 15:14:17.097954 | edb0044E.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_NEW_NAME CLOSE
2015-03-22 15:14:17.113554 | edbtmp.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-22 15:14:17.113554 | edb.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_NEW_NAME
2015-03-22 15:14:17.113554 | edb.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_NEW_NAME CLOSE
2015-03-22 15:14:18.174358 | edb.log | ARCHIVE NOT_CONTENT_INDEXED | RENAME_OLD_NAME

(venv)-(hainguyen@hainguyen)-[~/Lab]
$
(venv)-(hainguyen@hainguyen)-[~/Lab]
$ cat usn.csv | grep -i rename | wc -l
grep: (standard input): binary file matches
45

(venv)-(hainguyen@hainguyen)-[~/Lab]
$
```

```
(venv)-(hainguyen@hainguyen)-[~/Lab]
$ grep --text -P "\.jpg.*RENAME_OLD" usn.csv --color
2015-03-25 15:13:39.169804 | Chrysanthemum.jpg | NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-25 15:13:39.435003 | Desert.jpg | NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-25 15:13:39.544205 | Hydrangeas.jpg | NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-25 15:13:48.873005 | Jellyfish.jpg | NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-25 15:13:49.060204 | Koala.jpg | NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-25 15:13:49.216204 | Lighthouse.jpg | NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-25 15:13:49.356604 | Penguins.jpg | NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-25 15:13:49.465805 | Tulips.jpg | NOT_CONTENT_INDEXED | RENAME_OLD_NAME
2015-03-24 20:11:42.836174 | Chrysanthemum.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Desert.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Hydrangeas.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Jellyfish.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Koala.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Lighthouse.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Penguins.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Tulips.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.836174 | Chrysanthemum.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Desert.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Hydrangeas.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Jellyfish.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Koala.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Lighthouse.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Penguins.jpg | ARCHIVE | RENAME_OLD_NAME
2015-03-24 20:11:42.851774 | Tulips.jpg | ARCHIVE | RENAME_OLD_NAME
```

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "YOUR NAME Lab 6.pdf", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 6 From YOUR NAME**", replacing "YOUR NAME" with your real name.